NEUROSCIENCE

A brain-computer interface that evokes tactile sensations improves robotic arm control

Sharlene N. Flesher[1,2,3], John E. Downey[1,2,3,4], Jeffrey M. Weiss[1,5], Christopher L. Hughes[1,2,3], Angelica J. Herrera[1,2,3], Elizabeth C. Tyler-Kabara[6], Michael L. Boninger[1,2,5,7,8], Jennifer L. Collinger[1,2,3,5,8]†*, Robert A. Gaunt[1,2,3,5]†*

29 May 2020; accepted 25 March 2021
10.1126/science.abd0380

Presented on July 9th, 2021

Journal Club in CNBI Lab

Minsu Zhang

Introduction

1) Bidirectional BCI

2) Effect of ICMS on ARAT

3) Effect on object-transfer

Introduction

Motivation

- BCIs that with implanted electrodes can restore lost arm and hand function.
- Without tactile somatosensory feedback, manipulation tasks become clumsy and slow.

Methods

- Artificial tactile precepts were driven in real time in a robotic hand that responded to object contact and grasp force.
- This evoked intracortical microstimulation (ICMS) of somatosensory cortex, which were experienced as originating from person's own palm and fingers.

Results

- Bidirectional BCI that evokes tactile percepts substantially improves performance on functional tasks.
- Neural interface that mimics principles of sensorimotor control can be intuitively used by a person with extensive motor impairments.

Introduction

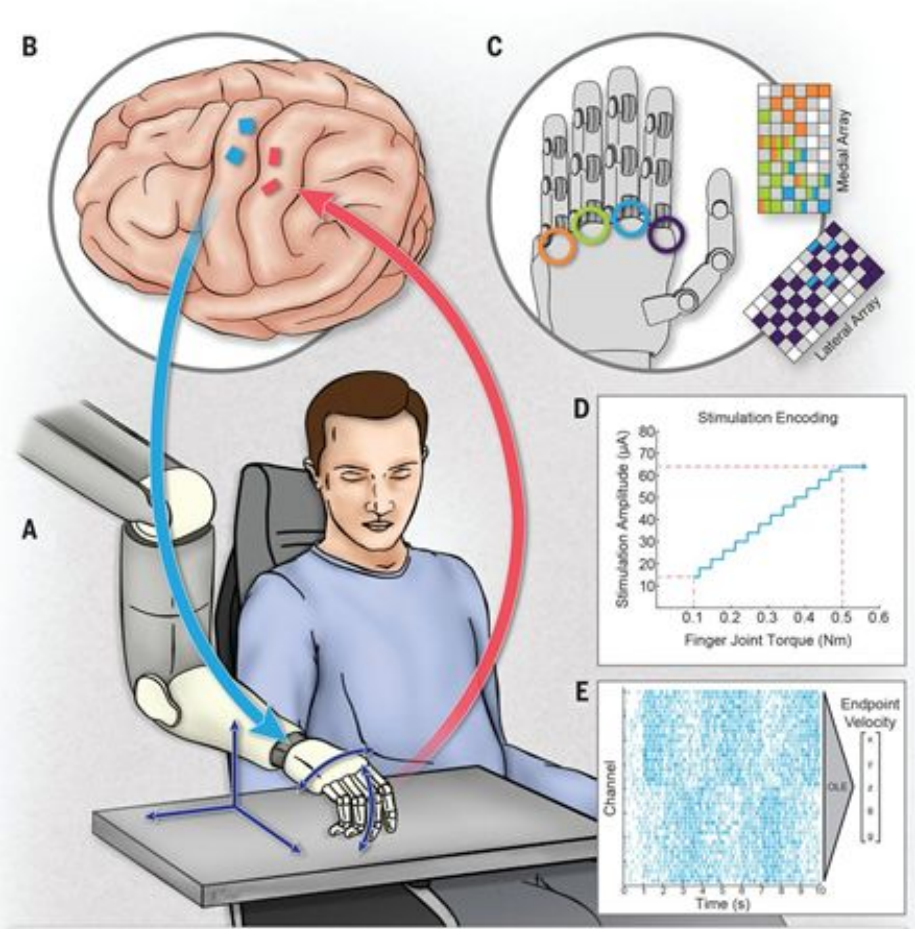
1) Bidirectional BCI

2) Effect of ICMS on ARAT

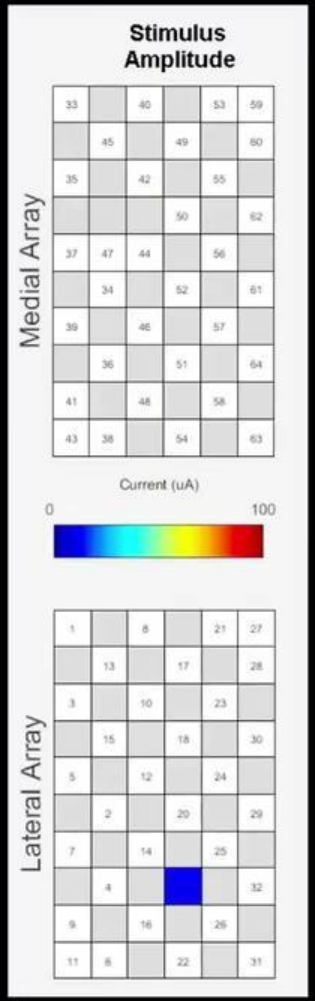
3) Effect on object-transfer

Conclusion

Bidirectional BCI system



- 2 x MEAs in motor cortex for decoding movement intent
- 2 x MEAs in cutaneous region of somatosensory cortex for palm & fingers sensation
- *Observation-based paradigm* to train 5 DoF velocity decoder
- Functional performance measured by Action Research Arm Test (ARAT)
  - The faster the higher score
  - Scores between 0 to 27



Torque on index finger triggers ICMS on corresponding electrode

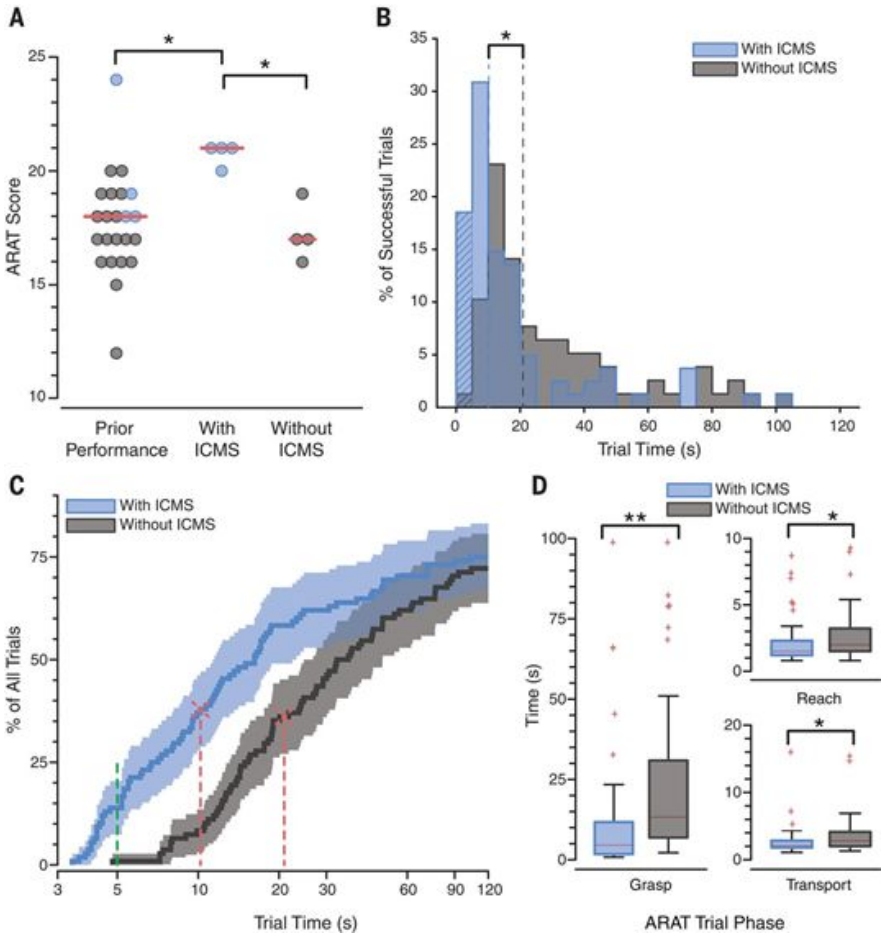
Introduction

1) Bidirectional BCI

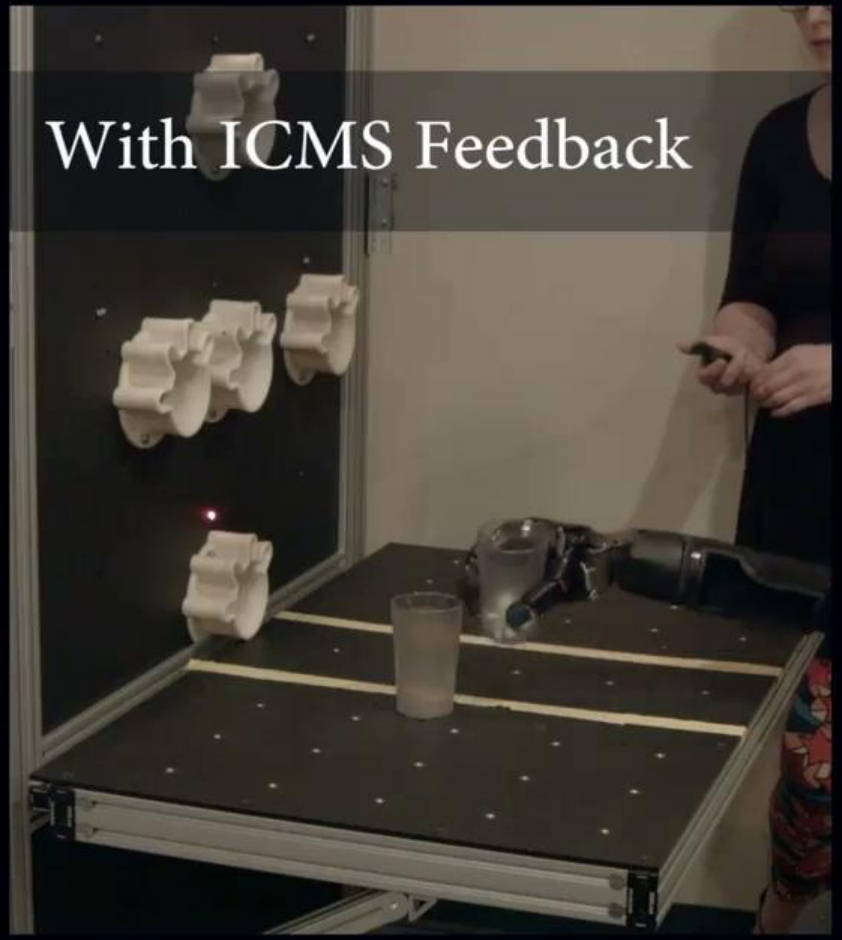
2) Effect of ICMS on ARAT

3) Effect on object-transfer

Effect of ICMS on ARAT

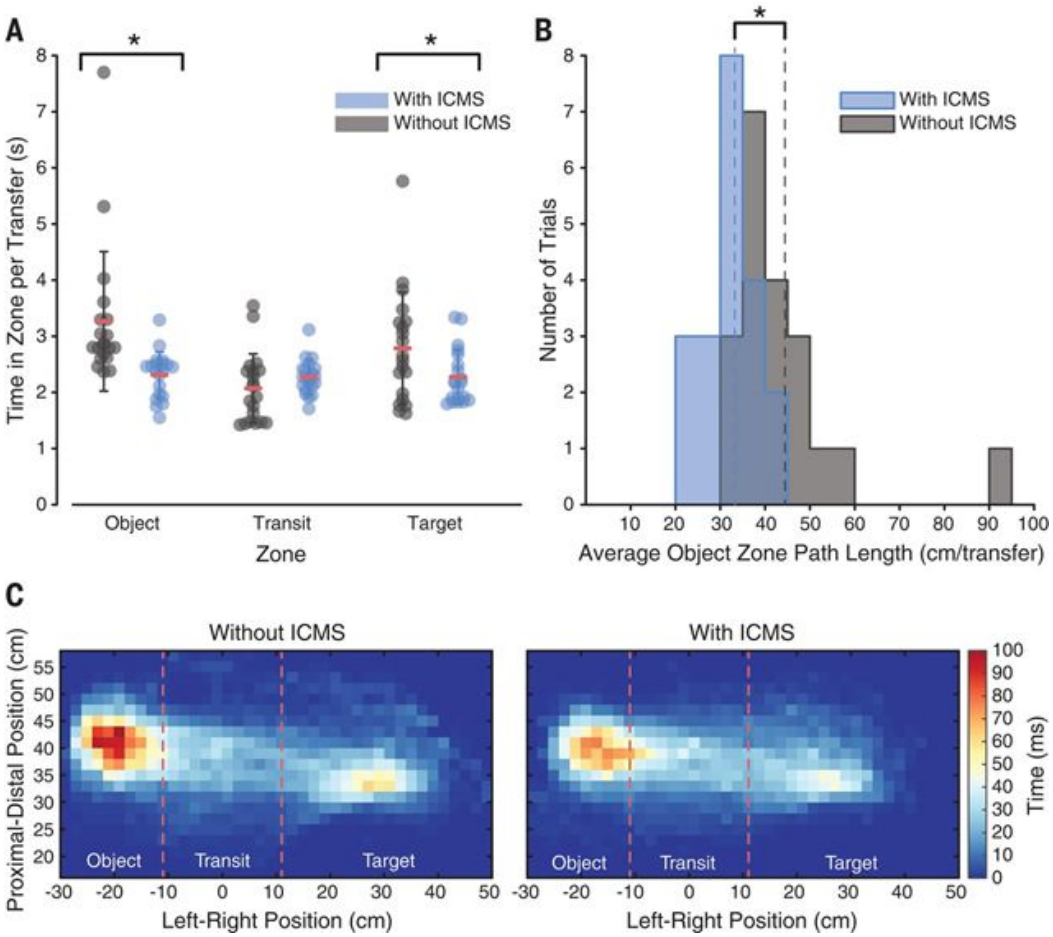


- (**B, C**) Faster completion times caused the increase in 3.5-point improvement in ARAT
- (**D**) Median time spent reaching and transport phase both decreased by 25% with ICMS



Fastest ARAT trials per feedback condition

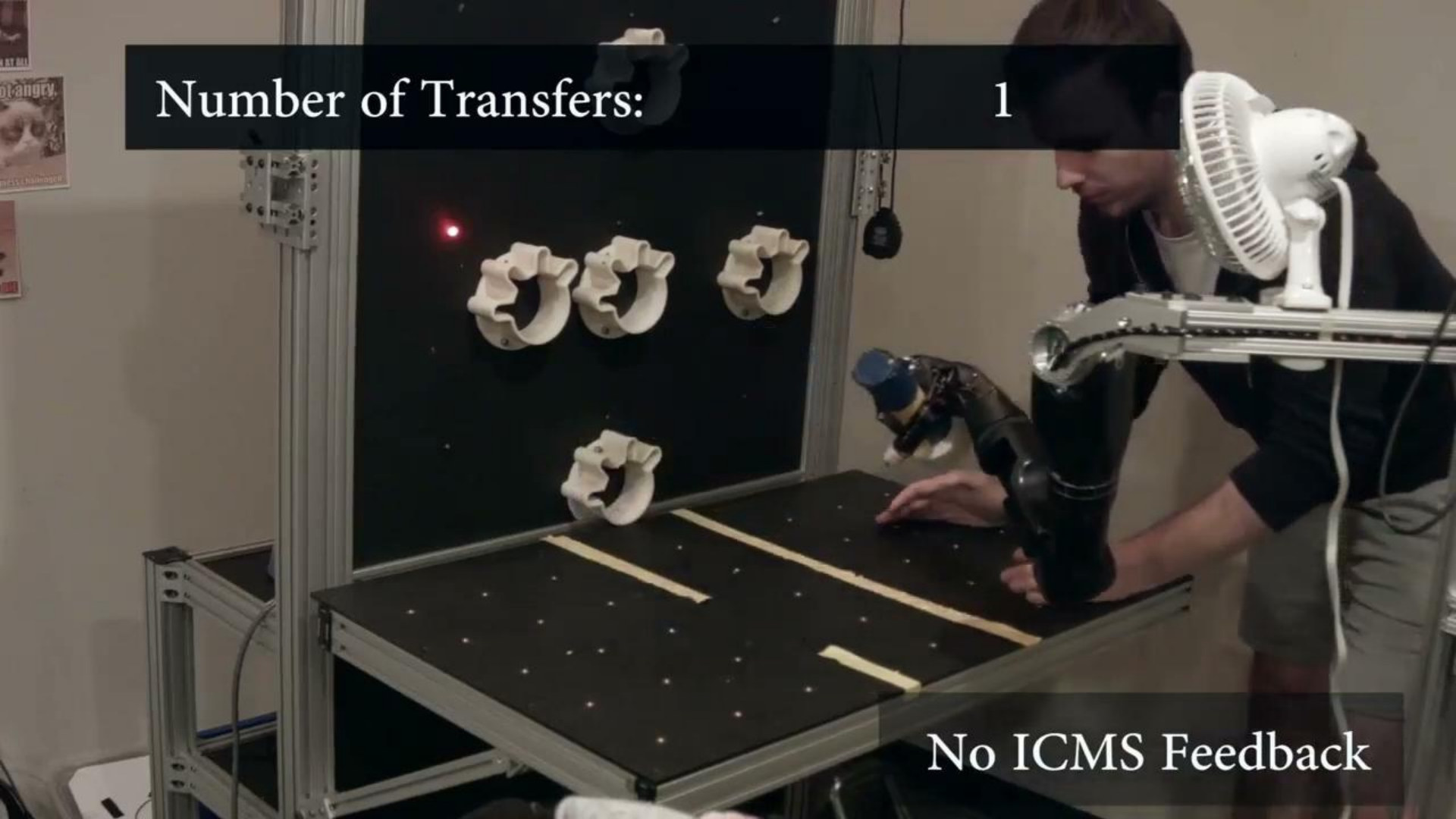
Effect of ICMS on object-transfer task



- Object transfer task:

  repeat max times in 2-min
  - Pick up (Object)
  - Carry (Transit)
  - Drop (Target)
- Time spent in the zones:
  - Object & Target

    ⇒ decreased
  - Transit ⇒ same


Number of Transfers: 1
No ICMS Feedback

References

1. Flesher, Sharlene N., et al. "A brain-computer interface that evokes tactile sensations improves robotic arm control." Science 372.6544 (2021): 831-836.
2. Osborn, Luke E., et al. "Prosthesis with neuromorphic multilayered e-dermis perceives touch and pain." Science robotics 3.19 (2018).